

Physical properties of ionic compounds

Learning outcomes

By the end of this section you should be able to explain the physical properties of ionic compounds in connection to its structure and bonding for the following:

- melting point
- volatility
- solubility in water
- electrical conductivity
- brittleness.

Ionic compounds are commonly found as salts, minerals and ceramics. Society typically uses the term salt to refer to sodium chloride, however, in chemistry this term is used to specify an ionic compound. While you might be most familiar with sodium chloride (table salt), salts exist in many different colours depending on the nature of the cation, anion or even the solvent within the lattice structure. Scroll through the images in **Interactive 1** to see some different examples of salts you may encounter in the course.

Interactive 1. Examples of Different Salts.

📄 More information for interactive 1

Thinking about table salt (NaCl), an ionic compound you are likely familiar with from everyday life, what are some properties you know about this salt? If you add it to a pot of water, will it dissolve? Does it melt easily? Does it break easily into

small pieces? Can it conduct electricity in its solid state? What about as a solution in water? Think about your experiences with table salt as we look more closely at the physical properties of ionic compounds.

Melting points

Think about all the examples of ionic compounds we have seen so far. What state of matter do you notice most of them have been in? Nearly every example of ionic compound we have looked at has been in its solid state. In [section S1.1.2a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/states-of-matter-and-changes-of-state-id-43067/>) you learned what change of state occurs during the process of melting. A melting point is the temperature at which a solid will undergo a change of state to a liquid. Would you expect substances that are solid at room temperature to have high or low melting points?

Melting an ionic compound to form a liquid involves breaking the electrostatic attractive forces between the oppositely charged ions in the lattice. Due to the strength and number of the electrostatic attractive forces between ions, a lot of energy is required to melt ionic compounds. Therefore, ionic compounds have high melting points. **Table 1** summarises some of the melting points for group 1 and 2 chlorides. Notice that these values are all quite high, much higher than can be achieved with an oven in your home!

Table 1. Melting points for group 1 and 2 chlorides.

Group 1 chlorides	Melting point (°C)	Group 2 chlorides	Melting point (°C)
LiCl	605	BeCl ₂	415
NaCl	801	MgCl ₂	714
KCl	770	CaCl ₂	772
RbCl	718	SrCl ₂	874
CsCl	645	BaCl ₂	962

The box below looks at a group of ionic compounds that have lower melting points than those typically referenced when discussing ionic compounds.

International Mindedness

Ionic liquids are a group of ionic compounds with significantly lower melting points (<100 °C). Ionic liquids are unique in that they can dissolve a large range of different solutes. Globally, many current solvents used in industry are flammable and give off volatile compounds that are harmful to the surrounding environment. Using ionic liquids as solvents reduces these concerns. Ionic liquids form an active area of research in Green Chemistry, an area of chemistry

focused on the minimisation and elimination of hazardous substances. Visit this [source](https://www.electrochem.org/dl/interface/spr/spr07/spr07_p38.pdf)  (https://www.electrochem.org/dl/interface/spr/spr07/spr07_p38.pdf) if you are interested in learning more about ionic liquids.

Volatility

The term volatility describes the ease at which a substance vaporises or becomes a gas. In [section S1.1.1b](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/separation-techniques-id-43066/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/separation-techniques-id-43066/>) you learned which changes of state are involved in volatility. Volatility would be the ease at which a substance evaporates, boils or sublimates. A substance with a high volatility easily becomes a gas. You have likely encountered volatile substances as the 'smelly' substances in vinegar, perfume and paints. A substance with a low volatility does not become a gas easily. As ionic compounds have high melting points and do not easily vaporise they have a low volatility.

There are, however, substances known as smelling salts. These salts are usually made of ammonium carbonate and were traditionally used to relieve faintness due to the odour given off. Recall what you learned about the strength of ionic bonds and lattice enthalpy in [section S2.1.3a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/>). How do you think the lattice enthalpy of smelling salts compares to that of typical salts?

Solubility

Have you ever noticed that when you shake table salt into water it dissolves? From [section S1.1.1a](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/elements-compounds-and-mixtures-id-43065/>), what is the name of the homogeneous mixture formed when table salt dissolves in water? When sodium chloride in table salt dissolves in water an aqueous solution is formed. Sodium chloride dissolves in water due to the strong interactions sodium and chloride ions have with water compared to between the oppositely charged ions in the lattice. These strong interactions allow water to surround individual ions, separating ions from their fixed positions in the lattice. **Figure 1** shows how a solution is formed from solid sodium chloride when it encounters water.

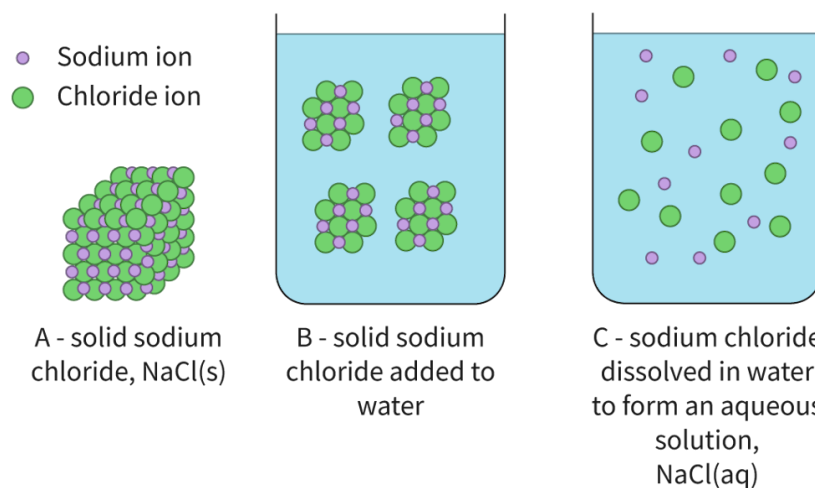
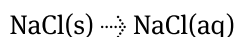
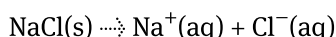


Figure 1. Solid sodium chloride dissolving in water. More information for figure 1

We can represent the physical change occurring during the formation of an aqueous solution for sodium chloride using state symbols as:



Since individual ions are surrounded by water in the aqueous solution of an ionic compound, we can also write the physical change occurring as:



Many, but not all, ionic compounds are soluble in water. The solubility of an ionic compound depends on the strength of the attraction between the ions and water along with the strength of the ionic bond. An ionic compound is insoluble in water when it does not form a homogeneous mixture when added to water.

If you try to shake table salt onto cooking oil, you might notice it does not dissolve. This is because there are no strong interactions between cooking oil and sodium and chloride ions. This makes sodium chloride insoluble in oil. What type of mixture is formed when an ionic compound is insoluble in a solvent?

Theory of Knowledge

Nacre, also known as mother of pearl, is produced on the inner shell of some oysters when a piece of foreign organic material, such as a parasite, enters the shell, or if the oyster's tissue is damaged by a predator. Nacre is composed of layers of calcium carbonate held together by organic material, producing a strong and beautifully iridescent substance highly valued in the art and jewellery industries. Historically, pearl oysters were collected by hand, opened and inspected for pearls — a very tedious practice as finding pearls is very rare and also wasteful, greatly depleting the pearl oyster population. Today, 95% of pearls available are 'cultured' by placing a small bead inside an oyster's shell and allowing the oyster to cover the bead with layers of nacre. The resulting pearl is produced faster and with a more consistent shape. This process, developed in 1956, also allowed the pearl oyster population to regenerate to a level that they no longer face extinction. Cultured pearls and natural pearls are almost identical in appearance — the only way to be absolutely certain is to examine the pearl by X-ray. Does the ability to replicate a natural process, producing a nearly identical product, reduce the beauty or value of nacre?

Conductivity

The term electrical conductivity refers to the ability of charged particles to move through a region of space. Recall what you know about ionic compounds. What charged particles are involved? Are these charged particles able to move? Can you think of situations where these charged particles can move?

Solid ionic compounds have cations and anions as charged particles that are fixed in the crystal lattice. This lattice structure in the solid state prevents movement of the charged ions. Therefore, solid ionic compounds are not electrically conductive.

When an ionic compound melts to form a molten liquid, the ionic bonds holding oppositely charged ions begin to break. This allows cations and anions to move throughout the liquid. Since there is motion of charged particles, molten ionic compounds are electrically conductive.

When an ionic compound dissolves in water to form an aqueous solution, ions are surrounded by water. Since ions are no longer fixed in positions (as they were in the solid lattice) they are free to move throughout the solution. As there is motion of charged particles, aqueous ionic compounds are electrically conductive. **Figure 2** illustrates the conductivity for ionic compounds in various states using an electrical circuit.

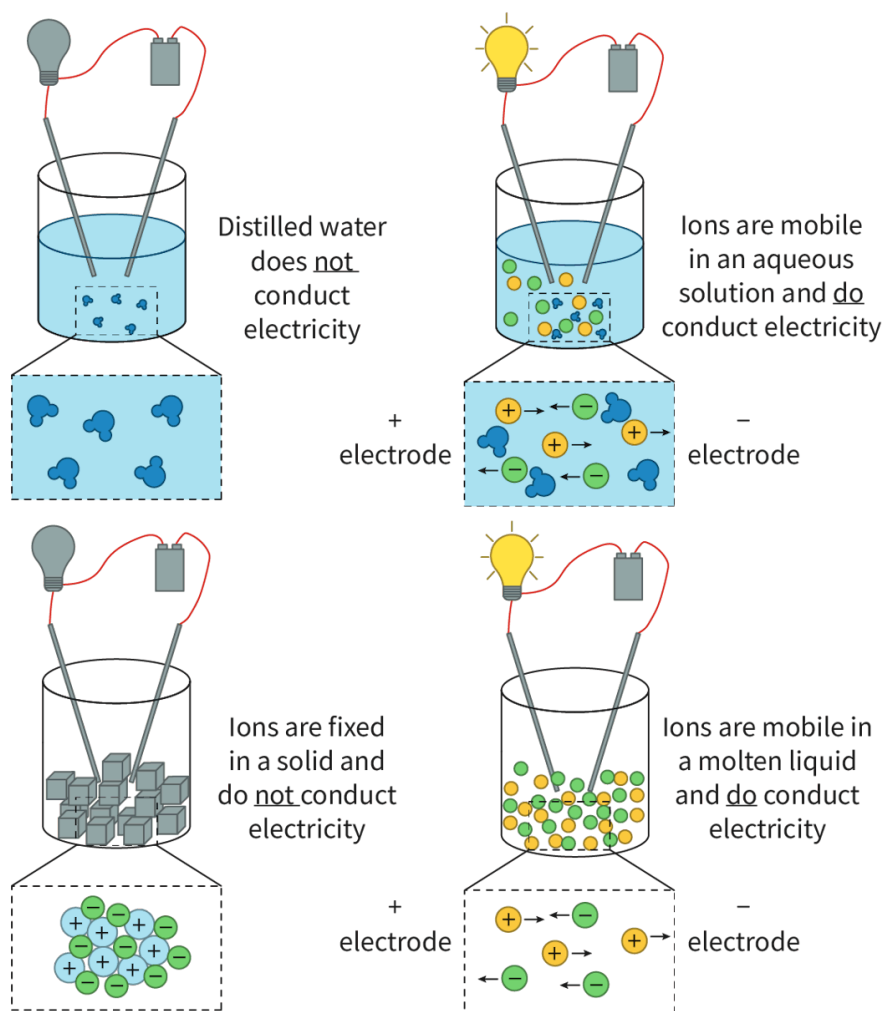


Figure 2. Electrical circuit showing when ionic compounds are electrically conductive.

© More information for figure 2

Watch **Video 1** to help you visualise the conductivity of ionic compounds in the states of matter listed above.

S2.1.3 Conductivity of Ionic Compounds [SL IB Chemistry]



Video 1. Visualise the conductivity of ionic compounds in the solid, molten and aqueous states.

The box below highlights common misconceptions of students when it comes to the conductivity of ionic compounds.

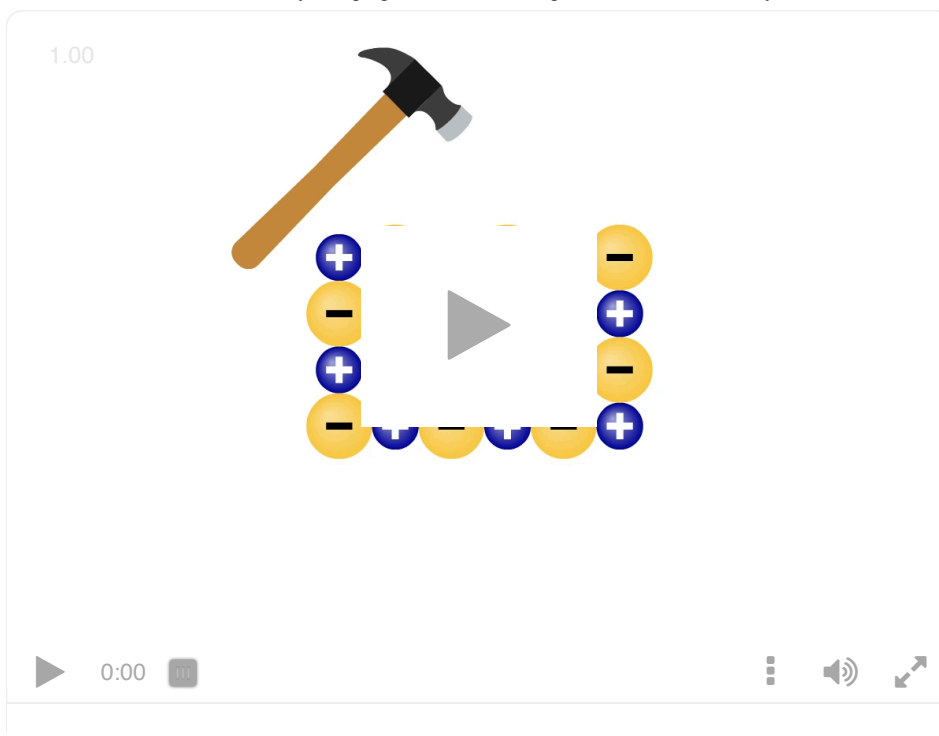
Study skills

Conductivity refers to the mobility of charged particles. It is very common for students to misinterpret conductivity to be exclusively due to electrons, however, in the case of aqueous ionic compounds, the conductivity of the solutions is from the mobility of the ions in solutions, **not** electrons.

Brittleness

Have you ever noticed if you have a large piece of table salt it can easily be ground into smaller pieces? This happens because sodium chloride is brittle. This is the final characteristic property of ionic compounds. While ionic bonds are strong, ionic substances fracture easily. These properties can be explained by taking a close look at the lattice structure as shown in **Interactive 2**.

When a stress is applied (caused by dropping or hitting with a hard object), layers within the lattice shift in response to that stress. As layers shift, ions become adjacent to like charges. These like charges experience an electrostatic repulsion and layers separate from one another, which forces the lattice to break into pieces. This is why ionic compounds, like sodium chloride in table salt, are brittle.



Interactive 2. Ionic Lattice Shifting in Response to an Applied Stress.

[More information for interactive 2](#)

The box below provides some examples of methods to collect qualitative and quantitative data for the properties of ionic compounds discussed in this section.

Practical skills

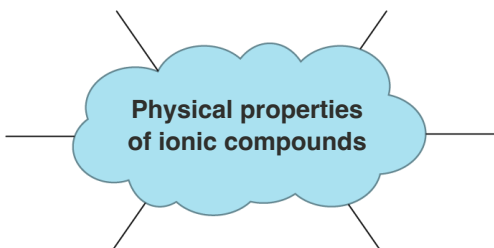
Inquiry 2: Collecting and processing data — Collecting data

There are qualitative and quantitative methods of collecting data to study the physical properties of ionic compounds. Below are examples of a few ways you could collect qualitative and quantitative data to study some of these physical properties.

	Qualitative data	Quantitative data
Melting point	Observe whether or not the substance melts when placed into the flame of a Bunsen burner.	Observe what temperature the substance melts at using a melting point apparatus.
Solubility in water	Observe whether a small amount of the solute forms a homogeneous mixture when added to water.	Use a conductivity meter to measure the conductivity of the solution compared to water. If the ionic compound is colourless, a spectrophotometer can also be used to measure the solubility.

	Qualitative data	Quantitative data
Electrical conductivity	Observe whether a lightbulb of a simple electrical circuit lights up.	Use a conductivity meter to measure conductivity of solution.
Brittleness	Observe (macroscopically with eyes or microscopically with microscope) the ease at which cracks and fractures form on the crystal's surface.	Use a structures and materials tester measures the force at which the material fractures.

Drag and drop the boxes in **Interactive 3** to see whether you can recall all the physical properties of ionic compounds.



Physical properties of ionic compounds

conducts electricity in solid state

low volatility

low melting points

none are soluble in water

conducts electricity in molten state

high volatility

high melting points

some are soluble in water

conducts electricity in aqueous state

brittle

flexible

all are soluble in water

☒ Check

Interactive 3. Summary of the Physical Properties of Ionic Compounds. More information for interactive 3**Creativity, activity, service****Strand:** Service**Learning outcomes:**

- Demonstrate how to initiate and plan a CAS experience.
- Demonstrate engagement with issues of global significance.

According to the [USGS](https://www.usgs.gov/faqs/why-ocean-salty#:~:text=The%20concentration%20of%20salt%20in,be%20about%20120%20million%20tons.)  (https://www.usgs.gov/faqs/why-ocean-salty#:~:text=The%20concentration%20of%20salt%20in,be%20about%20120%20million%20tons.), there are about 120 000 000 tons of salt in a cubic mile of seawater (which is equivalent to 1.09×10^{11} kg in 4.2 km^3 of seawater) and more than 90% of this salt is sodium chloride, an ionic compound. The world's oceans are a vast source of biodiversity but they are at risk from a variety of threats such as plastic pollution and overfishing.

8 June is [World Oceans Day](https://www.un.org/en/observances/oceans-day)  (https://www.un.org/en/observances/oceans-day) which aims to increase the public's perception of the threats posed to the world's oceans. What can you do at your school or in your local community to raise awareness of World Oceans Day?

In the following activity you will check your understanding of the physical properties of the ionic compounds covered in this section.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Thinking skills — Combining different ideas in order to create new understanding
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity for steps 1–5, pairs for step 6

1. Looking at the chemical formula for the substances below. Identify which ones you suspect are ionic. Explain how you made your selection.
 - (a) MgCl_2
 - (b) CO_2
 - (c) NH_4Br
 - (d) Na
2. Research the melting point of each of the substances above. Comment on how you might use the melting point to confirm or refute your selection above.
3. Explain why ionic compounds are brittle using a diagram to support your answer.

4. Sketch a diagram showing what you think happens at the atomic scale when sodium chloride dissolves in water.
5. Create a Venn diagram comparing solid ionic compounds to aqueous ionic compounds. Include structure, properties and diagrams.
6. Compare your work with a peer. Highlight three areas your work had in common. Highlight three areas where your work differed. Discuss these differences with your peer.